

Development tools and EDA—similarities and differences

A software development tool for embedded systems is an application or tool that is used in the design flow for designing, simulating, debugging, compiling, optimising and supporting these systems.

EDA tools are a category of such development tools. These include a schematic editor and simulator, signal integrity software and PCB design, layout and simulation. These, along with chip design, cross-compilation systems, in-circuit tools, debuggers, supervisory control and data acquisition (SCADA) and others, constitute software development tools that support embedded designs.

low sharing of files between users from around different parts of the world. Of course, a Web based system also comes with issues like the need for continuous high-speed Internet, security and increased uptime.

Fairchild's Power Supply Web-Designer (PSW), Texas Instruments' WEBENCH and the open source project Simulations for Performance Assessments that Report on Knowledge and Skills (SPARKS) are some of the online design and simulation tools currently available.

Altera and Mentor Graphics are collaborating to provide virtual platforms that contain simulation models of ARM processor sub-systems in Altera's systems on chip (SoC) field-programmable gate array (FPGA) families. This makes embedded software development easier while reducing the need for expensive hardware by enabling large development teams to perform development and testing in a virtual environment based on the chip.

Interconnected tools

Tools now encompass the entire system design flow. This means that something like wire harness tools now have the capability of being connected to their PCB tools, which are then combined and used for designing systems and even systems of systems.

Tools like the one Guo mentioned earlier in the article are also typically driven by knowledge based automation for every decision-making point of a design flow. This makes it possible for even novices and amateurs in the design field to complete a design quickly.

Engineers also want to know how they can do collision checking and can ensure that all ports fit perfectly on the circuit board, and soon.

This is why it became important to integrate the mechanical world into EDA tools. If the tool has support for Incremental Design Exchange format (IDX) files, it would help in the integration of design files from EDA tools to mechanical design tools like SolidWorks. The benefit here is that IDX allows incremental changes to the design file, in addition to making the file sizes smaller.

Tools for the IoT

Every other engineer we meet is into designing some product or the other for the IoT. So we thought it would be a good idea to have one section focusing entirely on the software tools available for those working on IoT products.

If you are trying to build M2M applications like remote monitoring or fleet management, M2MLabs Mainspring is a good framework that you could check out. Written in Java, it is an open source framework that allows flexible modelling of devices, their parts and characteristics. For instance, a temperature sensor can have a set of attributes related to indoor and outdoor data that it sends.

All data is time-stamped and sent according to a REST Web service. When new data is received, it triggers execution of Groovy scripts (programming language for Java platform) to perform boundary checking of data, interpolation or threshold supervision. Data is stored on scalablensSQL database.

DeviceHive and the IoT toolkit

are other frameworks that you could try out, too.

Node-RED is a tool for wiring together hardware devices and on-line services through different flows. It works with popular boards like Raspberry Pi (or Raspi), BeagleBone Black and Arduino. Flows could be created by adding an inject node and a debug node and then wiring the two together through a simple drag-and-drop GUI. You can then deploy the nodes. If you want to apply some function to each message from the inject node to the debug node, you can add a function node in between these—it is that simple!

Intel XDK IoT edition is a software tool that also uses Javascript to define the behaviour of an IoT device. It allows you to design a companion app I/O for the IoT device and control it from another smart device.

Wylidrin is a tool that Intel recommends. It can be used to create IoT projects using just your browser. Being a cloud based tool, it saves everything automatically, works in most browsers and is compatible with most popular boards. It allows boards to talk to each other, and enables mobile devices and Web services to talk to the board.

There are also neutral home automation software like OpenHAB that allow smart devices in your home to talk to each other. Being vendor-neutral, it allows users to easily add new features to their devices. To top these off, there are middleware like IoTSSyS and OpenIoT that ease the process of communicating data to and from devices.

Time to open up a new tool

We hope you have fun exploring these tools. *EFY Plus* subscribers would have had the chance to get most of these tools from the DVD accompanying their copy of the magazine over the last year. You can also check out in-depth reviews on development tools at www.electronicsforu.com/newelectronics/efydvd. Have fun! ●